

SRE CASE STUDY

AZURE ARC HYBRID CLOUD LAB

Weeks 1-3: Kubernetes, Azure Arc, Terraform, CI/CD, Ingress TLS, HPA, NetworkPolicy, RBAC

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GitHub: github.com/buth11/sre-homelab-azure-arc

Stack: Kubernetes | Azure Arc | Terraform | KQL | Azure DevOps | Traefik | HPA | NetworkPolicy | RBAC

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Chapter 1: Context and Architecture

This project demonstrates building a production-grade hybrid environment on GMKtec EVO-X2 (128 GB RAM) connecting Kubernetes with Microsoft Azure through Azure Arc. Three weeks of hands-on practice with documentation of every step and 10 documented incidents.

Layer	Technology	Purpose
Kubernetes	K3s v1.35.4+k3s1	3-node cluster
Ingress	Traefik + TLS	HTTPS domain routing
Autoscaling	HPA (autoscaling/v2)	CPU-based scaling 2-8 replicas
Network Security	NetworkPolicy	Intra-cluster firewall
Access Control	RBAC + ServiceAccount	Least privilege
Cloud Bridge	Azure Arc	Hybrid control plane
Monitoring	Azure Monitor + Prometheus	Observability
IaC	Terraform + Remote State	Infrastructure as Code
CI/CD	Azure DevOps	Pipeline with approval gate

Chapter 2: On-Premise Infrastructure

Node	IP	CPU	RAM
k3s-master (Control Plane)	192.168.122.10	4 vCPU	8 GB
k3s-worker1	192.168.122.11	4 vCPU	16 GB
k3s-worker2	192.168.122.12	4 vCPU	16 GB

Chapter 3: FinOps

The screenshot displays the Microsoft Azure Cost Management dashboard for user Bartosz Suszko. The interface is in Polish. On the left, a navigation pane lists various Azure services and tools. The main area is titled 'Cost Management: Bartosz Suszko | Budżety' (Budgets). It shows a table with one budget entry: 'SRE-Lab-Budget' with a budget of 5.00 USD and a current cost of 0.00 USD. The table columns include Name, Scope, Reset period, Creation date, Expiration date, Budget, Forecast, Estimated cost, and Progress. The progress bar for the budget is at 0.00%.

Nazwa	Zakres	Resetuj okres	Data utworzenia	Data wygaśnięcia	Budżet	Prognozowane	Oszacowane wy...	Postęp
SRE-Lab-Budget	55df28b8-39ed-58b...	Monthly	1.05.2026	30.04.2028	5.00 USD		\$0.00	0.00%

Screenshot 1: Azure Cost Management — 5 USD/month budget, cost: 0.00 USD.

Chapter 4: Azure Arc

The screenshot displays the Microsoft Azure portal interface. On the left, the navigation pane shows the 'Klastry Kubernetes' (Kubernetes Clusters) section under 'Podnieś poziom planu' (Raise the plan). The main content area shows the 'K3s-Homelab' cluster details. The cluster is connected and shows the following information:

- Podstawowe elementy** (Basic elements):
 - Subskrypcja (przenieś): [Azure subscription 1](#)
 - Identyfikator subskrypcji: b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
 - Grupa zasobów (przenieś): [SRE-Lab-RG](#)
 - Stan: Connected
 - Ustawienia: Lokalizacja: West Europe, Menedżer floty: Kliknij tutaj, aby przypisać.
- Właściwości** (Properties):
 - Wersja agenta: 1.33.0
 - Czas wygaśnięcia certyfikatu tożsamości zarządzanej: August 19, 2026 at 08:20 AM GMT+2
 - Wersja platformy Kubernetes: 1.35.4+k3s1
 - Łączna liczba węzłów: 3
 - Łączna liczba rdzeni: 12

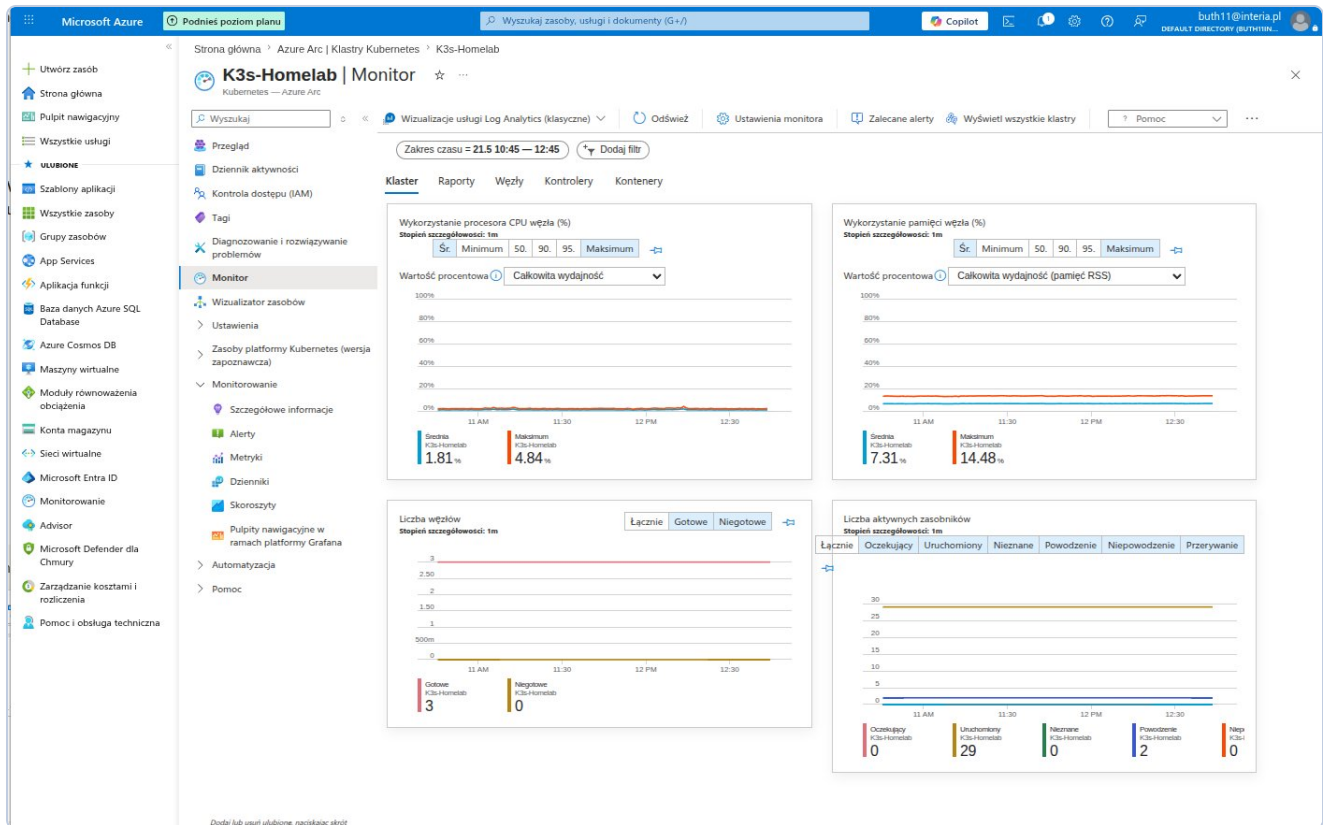
The right sidebar shows the 'Monitorowanie' (Monitoring) tab, which includes a 'Widok kodu JSON' (JSON View) button and a 'Rozszerzenia' (Extensions) section with 'azuremonitor-containers'.

Screenshot 2: Azure Arc — K3s-Homelab Connected, v1.35.4+k3s1, 3 nodes, 12 cores.

Chapter 5: Observability

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
{
  "principalId": "a3fe4a8c-7a2e-4b99-8de9-2150d617cb76",
  "tenantId": null,
  "type": "SystemAssigned"
},
"isSystemExtension": false,
"name": "azuremonitor-containers",
"packageUri": null,
"plan": null,
"provisioningState": "Succeeded",
"releaseTrain": "Stable",
"resourceGroup": "SRE-Lab-RG",
"scope": {
  "cluster": {
    "releaseNamespace": "azuremonitor-containers"
  },
  "namespace": null
},
"statuses": [],
"systemData": {
  "createdAt": "2026-05-22T07:44:41.936626+00:00",
  "createdBy": null,
  "createdByType": null,
  "lastModifiedAt": "2026-05-22T07:44:41.936626+00:00",
  "lastModifiedBy": null,
  "lastModifiedByType": null
},
"type": "Microsoft.KubernetesConfiguration/extensions",
"version": null
}
buth11@k3s-master:~$ sudo k3s kubectl get pods -n kube-system | grep ama
[sudo] password for buth11:
ama-logs-9m9x9          3/3      Running    0          37s
ama-logs-9scpr          3/3      Running    0          37s
ama-logs-rs-65b58c7f96-f8lds 2/2      Running    0          37s
ama-logs-spcdp          3/3      Running    0          37s
buth11@k3s-master:~$
```

Screenshot 3: AMA pods Running — 4 agents, 0 restarts.



Screenshot 4: Azure Monitor — CPU avg 1.58%, RAM avg 6.27%, 3 nodes Ready, 29 pods.

Chapter 6: KQL

The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with 'Log Analytics' selected. The main area shows the 'Logs' section of a Log Analytics workspace. A KQL query is entered in the 'New Query 1' editor:

```
1 KubeNodeInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Computer
4 project
5   Computer,
6   Status,
7   KubeletVersion
8   order by Computer asc
```

The query results are displayed in a table with columns: Computer, Status, and KubeletVersion. The results show 3 nodes, all in 'Ready' status, with 'v1.35.4+k3s1' version.

Computer	Status	KubeletVersion
k3s-master	Ready	v1.35.4+k3s1
k3s-worker1	Ready	v1.35.4+k3s1
k3s-worker2	Ready	v1.35.4+k3s1

KQL-1: 3 nodes Ready, v1.35.4+k3s1.

DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU | Logs

Log Analytics workspace

New Query 1* ... +

Time range: Set in query Show: 1000 results KQL mode

```

1 KubePodInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Name, Namespace
4 summarize PodCount = count() by Namespace, PodStatus
5 order by Namespace asc, PodCount desc

```

Namespace	PodStatus	PodCount
azure-arc	Running	12
default	Running	3
kube-system	Running	14
kube-system	Succeeded	2

0s 533ms Display time (UTC+00:00) Query details 1 - 4 of 4

KQL-2: azure-arc 12 Running, default 3 Running, kube-system 14+2.

DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU | Logs

Log Analytics workspace

New Query 1* ... +

Time range: Set in query Show: 1000 results KQL mode

```

1 KubeEvents
2 where TimeGenerated > ago(24h)
3 project TimeGenerated, Namespace, Name, Reason, Message
4 order by TimeGenerated desc
5 take 20

```

TimeGenerated [UTC]	Namespace	Name	Reason
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-9cpr	FailedMount
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-rs-65b58c7f96-f8ld	FailedMount
5/22/2026, 7:35:32.000 AM	azure-arc	kube-aad-proxy-5f59465457-12...	Unhealthy
5/22/2026, 7:35:25.000 AM	azure-arc	config-agent-7548f59c5-k4zrv	Unhealthy
5/22/2026, 7:35:13.000 AM	k3s-worker2		Rebooted
5/22/2026, 7:35:11.000 AM	k3s-worker1		Rebooted
5/22/2026, 7:35:09.000 AM	kube-system	metrics-server-786d997795-zz...	Unhealthy
5/22/2026, 7:35:07.000 AM	k3s-master		Rebooted
5/21/2026, 10:53:39.000 AM	kube-system	ama-logs-89tpj	Unhealthy
5/21/2026, 10:52:38.000 AM	kube-system	ama-logs-rs-65b58c7f96-nkshw	Unhealthy
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling

0s 287ms Display time (UTC+00:00) Query details 1 - 13 of 13

KQL-3: 13 events 24h — Rebooted, Unhealthy (temporary), FailedScheduling.

The screenshot shows the Microsoft Azure Log Analytics interface. The workspace is 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. A new KQL query is being executed, showing results for pod restarts. The query filters for pods with a restart count greater than 0 and orders them by restart count in descending order.

```

1 KubePodInventory
2 where TimeGenerated > ago(24h)
3 summarize arg_max(TimeGenerated, *) by Name, Namespace
4 where PodRestartCount > 0
5 project Namespace, Name, PodRestartCount, PodStatus
6 order by PodRestartCount desc

```

Namespace	Name	PodRestartCount	PodStatus
azure-arc	extension-manager-85899588c...	3	Running
azure-arc	clusterconnect-agent-5bc68597...	3	Running
kube-system	svcib-traefik-4b75cb6-8b8x...	2	Running
azure-arc	extension-events-collector-6c9...	2	Running
azure-arc	clusteridentityoperator-767768...	2	Running
azure-arc	kube-aad-proxy-5f5465457-42...	2	Running
kube-system	svcib-traefik-4b75cb6-c2gzs...	2	Running
azure-arc	config-agent-75548f59c5-k4zrv...	2	Running
azure-arc	cluster-metadata-operator-6db...	2	Running
kube-system	svcib-traefik-4b75cb6-9587...	2	Running
azure-arc	metrics-agent-6d9f94764-rc44...	2	Running
azure-arc	controller-manager-7cb948ddff...	2	Running
azure-arc	resource-sync-agent-5f4688cc5...	2	Running
azure-arc	flux-logs-agent-76f8c454c9-9g...	1	Running
azure-arc	logcollector-8bd4468b4-4s2zw...	1	Running
kube-system	ama-logs-nlg5p...	1	Running
kube-system	helm-install-traefik-ggts...	1	Succeeded
kube-system	svcib-aras-demo-app-8581d79...	1	Running
default	aras-demo-app-f78659546-vb...	1	Running
kube-system	coredns-c4dbfbb5f-llkx...	1	Running
kube-system	local-path-provisioner-5c4dc5d...	1	Running
kube-system	metrics-server-786d997795-zz...	1	Running
kube-system	svcib-aras-demo-app-8581d79...	1	Running

KQL-4: 27 pods with restarts after VM reboot — all Running.

The screenshot shows the Microsoft Azure Log Analytics interface. The workspace is 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. A new KQL query is being executed, showing results for SLO 100% over the last 24 hours. The query calculates the availability percentage based on ready samples and total samples.

```

1 KubeNodeInventory
2 where TimeGenerated > ago(24h)
3 summarize
4   TotalSamples = count(),
5   ReadySamples = countif(Status == "Ready")
6   by Computer
7 extend AvailabilityPct = round((toReal(ReadySamples) / toReal(TotalSamples)) * 100, 2)
8 project Computer, AvailabilityPct, ReadySamples, TotalSamples
9 order by Computer asc

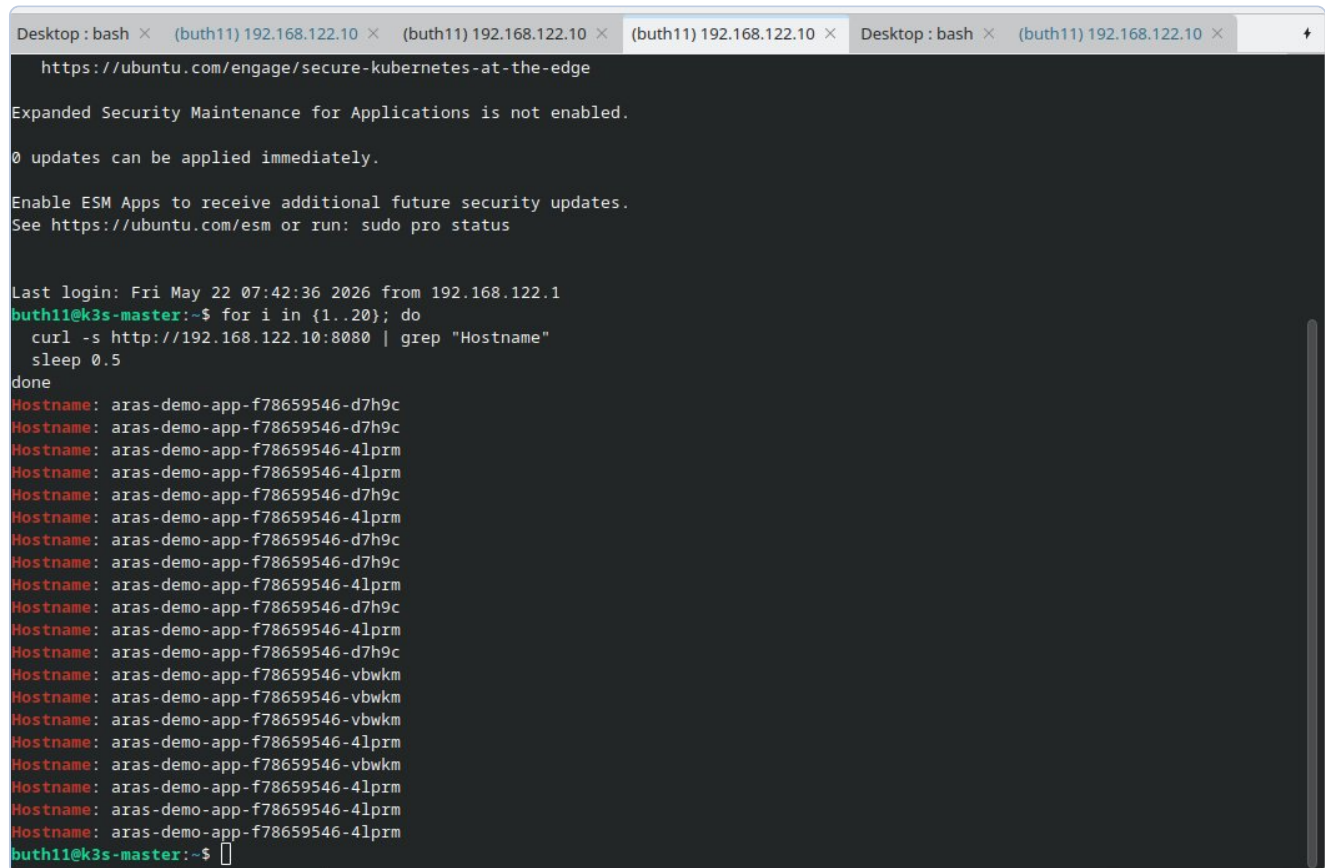
```

Computer	AvailabilityPct	ReadySamples	TotalSamples
k3s-master	100	184	184
k3s-worker1	100	184	184
k3s-worker2	100	184	184

KQL-5: SLO 100% — 184/184 samples Ready.

SLO 100% in 24h window. Error budget intact even after full VM restart.

Chapter 7: Load Balancing



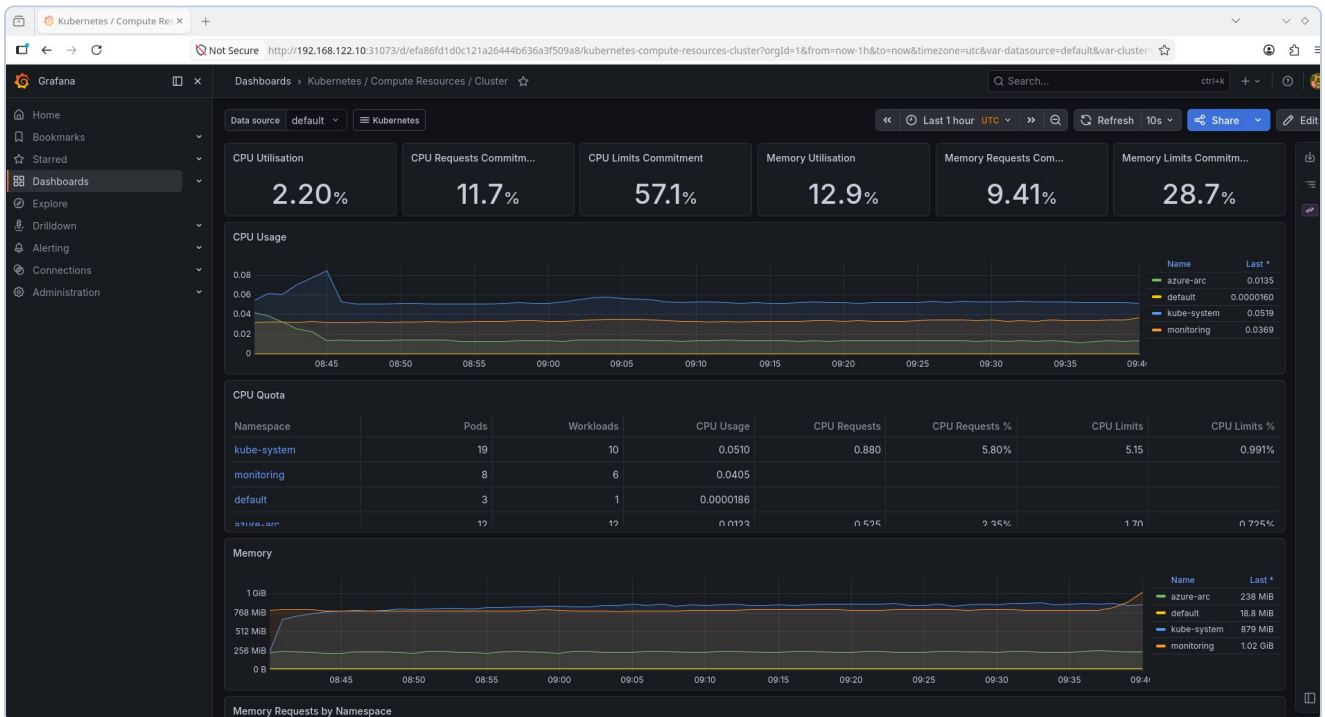
The screenshot shows a terminal window with multiple tabs. The active tab is titled "(buth11) 192.168.122.10". The terminal output includes a security update notification, a login message, and the execution of a loop that sends 20 curl requests to a specific URL. The output of these requests shows a mix of hostnames, indicating a round-robin load balancing distribution across three pods.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May 22 07:42:36 2026 from 192.168.122.1
buth11@k3s-master:~$ for i in {1..20}; do
  curl -s http://192.168.122.10:8080 | grep "Hostname"
  sleep 0.5
done
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
buth11@k3s-master:~$
```

Screenshot 5: Round-robin — 20 requests across 3 pods, 900 parallel with zero errors.

Chapter 8: Prometheus & Grafana



Screenshot 6: Grafana — CPU 2.20%, RAM 12.9%, CPU Quota per namespace.

Chapter 9: Terraform IaC

```
Desktop : bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform : bash × (buth11) 192.168.122.10 ×
Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ action_group_id           = (known after apply)
+ budget_name               = "SRE-Lab-Budget"
+ log_analytics_workspace_id = (known after apply)
+ log_analytics_workspace_key = (sensitive value)
+ resource_group_id         = (known after apply)
+ resource_group_name        = "SRE-Lab-RG"
+ subscription_id           = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"

Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
  terraform apply "tfplan"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ terraform apply tfplan
random_string.suffix: Creating...
random_string.suffix: Creation complete after 0s [id=5plwbb]
azurerm_resource_group.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Still creating... [00m10s elapsed]
azurerm_consumption_budget_subscription.sre_lab: Creation complete after 19s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]

Error: A resource with the ID "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG" already exists -
to be managed via Terraform this resource needs to be imported into the State. Please see the resource documentation for "azure
rm_resource_group" for more information.

with azurerm_resource_group.sre_lab,
on main.tf line 26, in resource "azurerm_resource_group" "sre_lab":
26: resource "azurerm_resource_group" "sre_lab" {

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Plan: 6 resources, 0 changes.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×

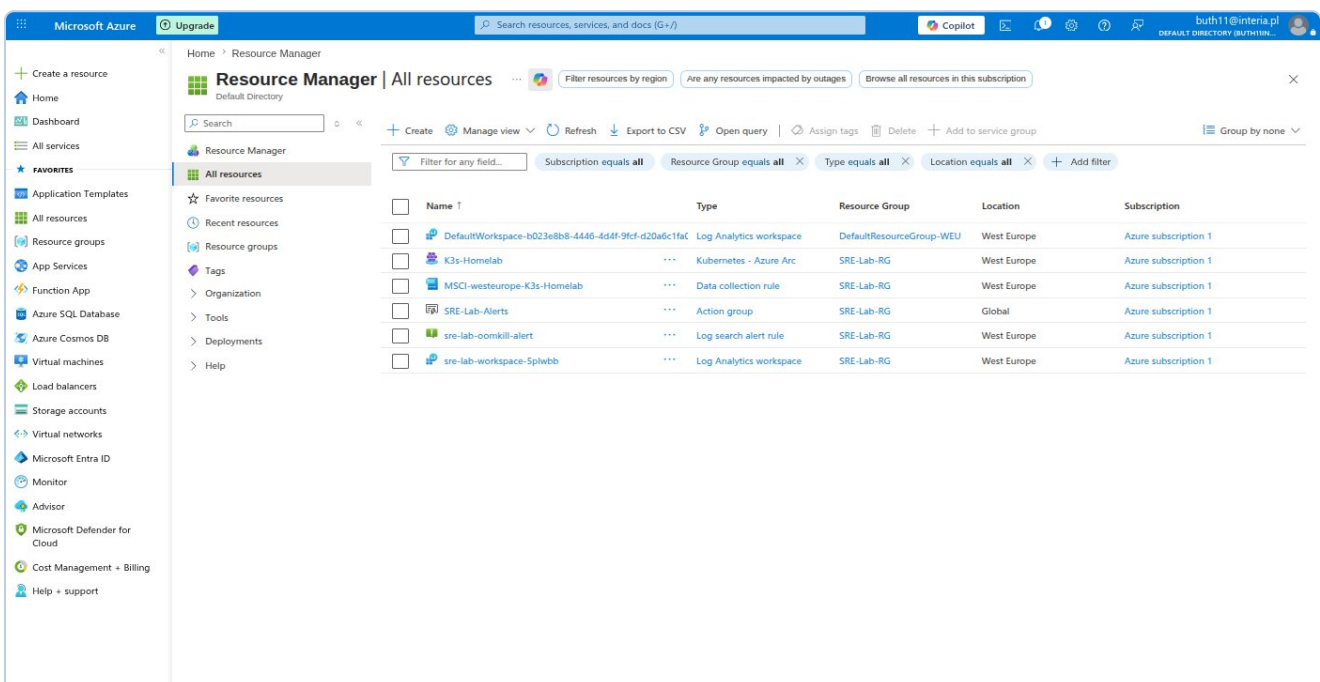
Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
terraform apply "tfplan"
azurerm_resource_group.sre_lab: Modifying... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_resource_group.sre_lab: Modifications complete after 3s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_monitor_action_group.sre_alerts: Creating...
azurerm_log_analytics_workspace.sre_lab: Creating...
azurerm_monitor_action_group.sre_alerts: Creation complete after 4s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.sre_lab: Creation complete after 47s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creating...
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creation complete after 5s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

Apply complete! Resources: 3 added, 1 changed, 0 destroyed.

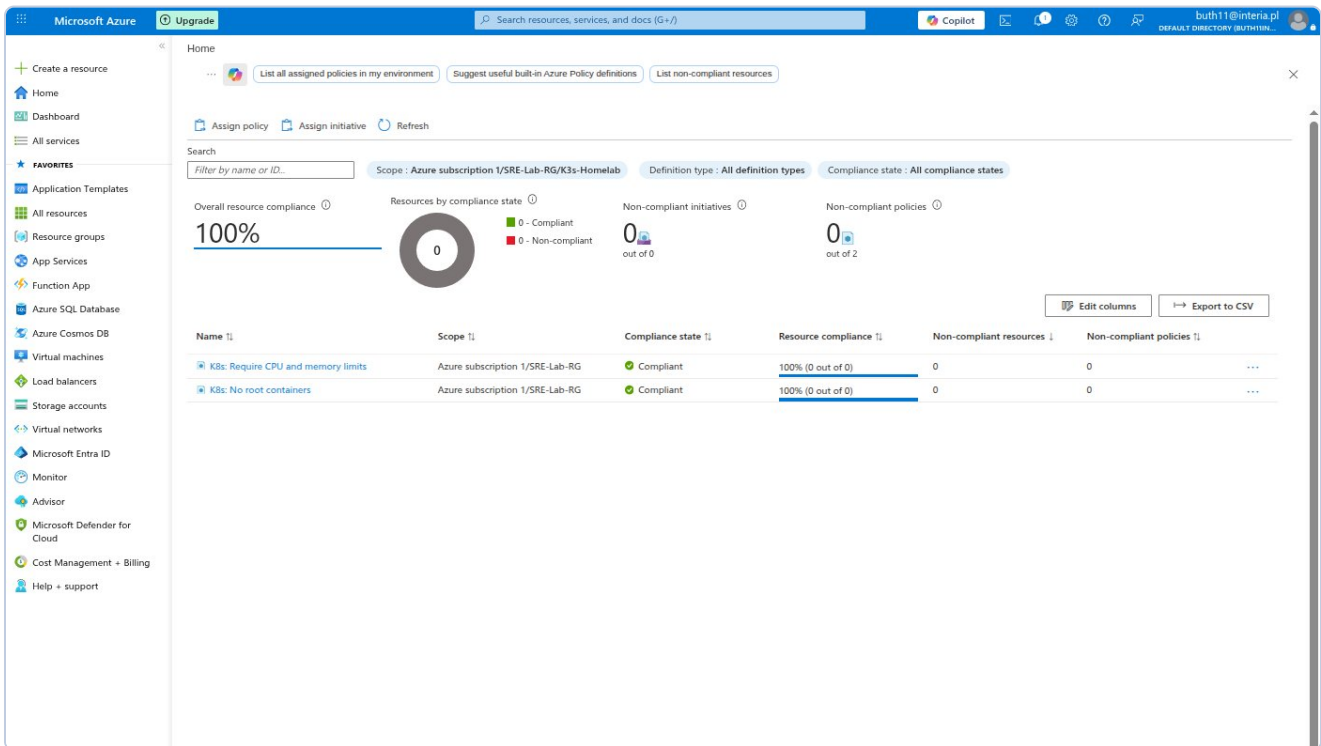
Outputs:
action_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts"
budget_name = "SRE-Lab-Budget"
log_analytics_workspace_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb"
log_analytics_workspace_key = <sensitive>
resource_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG"
resource_group_name = "SRE-Lab-RG"
subscription_id = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Apply: 3 added, 1 changed.



Azure Portal: Terraform resources in SRE-Lab-RG.

Chapter 10: Azure Policy



Azure Policy: No root containers + CPU/memory limits — both Compliant.

Chapter 11: GitHub

The screenshot shows the GitHub repository page for 'buth11 / sre-homelab-azure-arc'. The repository is public and has 1 branch (main) and 0 tags. The file list shows the initial commit: 'K3s + Azure Arc SRE Homelab' from 8 minutes ago. The files listed are: docs, k8s, monitoring, scripts, terraform, .gitignore, and README.md. The README file is selected, showing the title 'SRE Homelab: Azure Arc Hybrid Cloud Lab' and a description: 'A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.' The right sidebar shows the repository's statistics: 0 stars, 0 watching, and 0 forks. The 'About' section is empty. The 'Releases' section shows no releases published. The 'Packages' section shows no packages published. The 'Contributors' section shows 1 contributor: buth11. The 'Languages' section shows a bar chart with Shell at 58.4% and HCL at 41.6%.

File	Commit	Time
docs	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
k8s	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
monitoring	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
scripts	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
terraform	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
.gitignore	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
README.md	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago

README

SRE Homelab: Azure Arc Hybrid Cloud Lab

A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.

Overview

Statistics:

- Stars: 0
- Watching: 0
- Forks: 0

Releases: No releases published. [Create a new release](#)

Packages: No packages published. [Publish your first package](#)

Contributors: 1 contributor: buth11

Languages:

Language	Percentage
Shell	58.4%
HCL	41.6%

GitHub repo sre-homelab-azure-arc — public, Shell 58.4%, HCL 41.6%.

Chapter 12: Week 2 — Service Principal, Remote State, CI/CD

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # Czeka 30 sekund na propagacje uprawnień w Azure AD
sleep 30

az storage blob list \
  --container-name tfstate \
  --account-name sretfstate5496 \
  --auth-mode login \
  --query "[].{name:name, size:properties.contentLength, modified:properties.lastModified}" \
  --output table

You do not have the required permissions needed to perform this operation.
Depending on your operation, you may need to be assigned one of the following roles:
  "Storage Blob Data Owner"
  "Storage Blob Data Contributor"
  "Storage Blob Data Reader"
  "Storage Queue Data Contributor"
  "Storage Queue Data Reader"
  "Storage Table Data Contributor"
  "Storage Table Data Reader"

If you want to use the old authentication method and allow querying for the right account key, please use the "--auth-mode" parameter and "key" value.

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # Pobiera klucz dostępu do Storage Account
# auth-mode key używa klucza zamiast OAuth - omija problem z uprawnieniami SP
STORAGE_KEY=$(az storage account keys list \
  --account-name sretfstate5496 \
  --resource-group terraform-state-rg \
  --query "[0].value" -o tsv)

# Listuje pliki w kontenerze używając klucza
az storage blob list \
  --container-name tfstate \
  --account-name sretfstate5496 \
  --account-key $STORAGE_KEY \
  --query "[].{name:name, size:properties.contentLength, modified:properties.lastModified}" \
  --output table
Name                Size      Modified
-----
sre-homelab.tfstate 14881     2026-05-23T18:23:07+00:00
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Screenshot W2-1: Remote State — sre-homelab.tfstate in tfstate container.

Microsoft Azure Upgrade Search resources, services, and docs (G+/I) buth11@interia.pl DEFAULT DIRECTORY (BUTH11IN...

Create a resource

Home Dashboard All services FAVORITES Application Templates All resources Resource groups App Services Function App Azure SQL Database Azure Cosmos DB Virtual machines Load balancers Storage accounts Virtual networks Microsoft Entra ID Monitor Advisor Microsoft Defender for Cloud Cost Management + Billing Help + support

sre-homelab.tfstate

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL	https://sretfstate5496.blob...
LAST MODIFIED	5/23/2026, 8:23:07 PM
CREATION TIME	5/23/2026, 8:23:07 PM
VERSION ID	2026-05-23T18:23:07.9861361Z
TYPE	Block blob
SIZE	14.53 KiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DEB8F856BD8961
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/json
CONTENT-MD5	2mHfbTY53lmCzqo/RW5+MQ...
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	
LEASE STATUS	Unlocked
LEASE STATE	Available
LEASE DURATION	-
COPY STATUS	-
COPY COMPLETION TIME	-

Undelete

Metadata

Key	Value

Blob index tags

Key	Value

Screenshot W2-2: Azure Portal — blob properties: 14.53 KiB, encrypted, versioning active.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # terraform plan pobierze state z Azure Blob zamiast lokalnego pliku
# Powinno pokazać "No changes" bo infrastruktura jest aktualna
terraform plan
Acquiring state lock. This may take a few moments...
random_string.suffix: Refreshing state... [id=5plwbb]
data.azure_rm_subscription.current: Reading...
azure_rm_resource_group.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azure_rm_monitor_action_group.sre_alerts: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azure_rm_log_analytics_workspace.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
data.azure_rm_subscription.current: Read complete after 1s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093]
azure_rm_consumption_budget_subscription.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]
azure_rm_monitor_scheduled_query_rules_alert_v2.oom_kill: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
Releasing state lock. This may take a few moments...
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Screenshot W2-3: terraform plan — Acquiring/Releasing state lock. No changes.

The screenshot shows the Azure DevOps web interface. On the left is a sidebar with navigation options: Overview, Boards, Repos, Pipelines, Environments, Releases, Library (selected), Task groups, Deployment groups, Test Plans, and Artifacts. The main area is titled 'Library > terraform-credentials'. Below this, there are tabs for 'Variable group', 'Save', 'Clone', 'Security', 'Pipeline permissions', 'Approvals and checks', and 'Help'. The 'Properties' section shows the 'Variable group name' as 'terraform-credentials' and a description field. There is a toggle switch for 'Link secrets from an Azure key vault as variables' which is currently off. The 'Variables' section contains a table with the following data:

Name ↑	Value	Icon
ARM_CLIENT_ID	54c7e017-a745-4e1e-b9da-d74f4901a3de	
ARM_CLIENT_SECRET	*****	🔒
ARM_SUBSCRIPTION_ID	b023e8b8-4446-4d4f-9fcf-d20a6c1fa093	
ARM_TENANT_ID	fd7288c4-e003-40fd-b3cf-9c0d1821cf5d	

At the bottom of the variables list is a '+ Add' button. The bottom of the sidebar shows 'Project settings' with a double arrow icon.

Screenshot W2-4: Azure DevOps Variable Group — ARM_* with hidden CLIENT_SECRET.

Azure DevOps

buth11

/ sre-homelab-arc

/ Pipelines

/ Environments

/ production

Search

BS

sre-homelab-arc

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Releases

Library

Task groups

Deployment groups

Test Plans

Artifacts

Project settings

<<

production

Automatically created environment

Add resource

Deployments

Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel.
[Learn more](#)

+ Add new

Order	Display name	Type	Timeout	Details
1	All approvers must approve	Approvals	30d	8%

Screenshot W2-5: Environment production — Approval Gate active.

Azure DevOps | sre-homelab-arc | Pipelines | buth11.sre-homelab-azure-... | 20260523.1

Overview
Boards
Repos
Pipelines
Pipelines
Environments
Releases
Library
Task groups
Deployment groups
Test Plans
Artifacts

Project settings

Jobs in run #20260523.1

buth11.sre-homelab-azure-arc

Terraform Validate & Plan

terraform plan	34s
Initialize job	<1s
Checkout buth1...	2s
Install Terraform...	2s
terraform init	7s
terraform validate	1s
terraform plan	15s
Publish terrafor...	3s
Post-job: Chec...	<1s
Finalize Job	<1s

Terraform Apply

terraform apply

terraform plan

View raw log

```
1 Agent: Azure Pipelines 1
2 Started: Just now
3 Duration: 34s
4
5 Job preparation parameters
44 1 artifact produced
45 Job live console data:
46 Starting: terraform plan
47 Async Command Start: DetectDockerContainer
48 Async Command End: DetectDockerContainer
49 Async Command Start: DetectDockerContainer
50 Async Command End: DetectDockerContainer
51 Finishing: terraform plan
```

Screenshot W2-6: Stage 1 — all steps green, 1 artifact.

Azure DevOps | sre-homelab-arc | Pipelines | #20260523.1 • feat: add Azure DevOps CI/CD pipeline for Terraform

Run new

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary | Environments | Associated pipelines | Code Coverage

Manually run by Bartosz Suszko
Today at 9:13 PM 4m 24s

Repository and version: sre-homelab-arc / main e9da4bfb

Related: 0 work items, 1 published

Tests and coverage: Get started

View change | Parameters

Stages | Jobs

Terraform Validate ...
1 job completed 37s
1 artifact

Terraform Apply
1 job completed 49s
1 checks passed

Project settings

Screenshot W2-7: Pipeline complete — both stages green, 4m 24s.

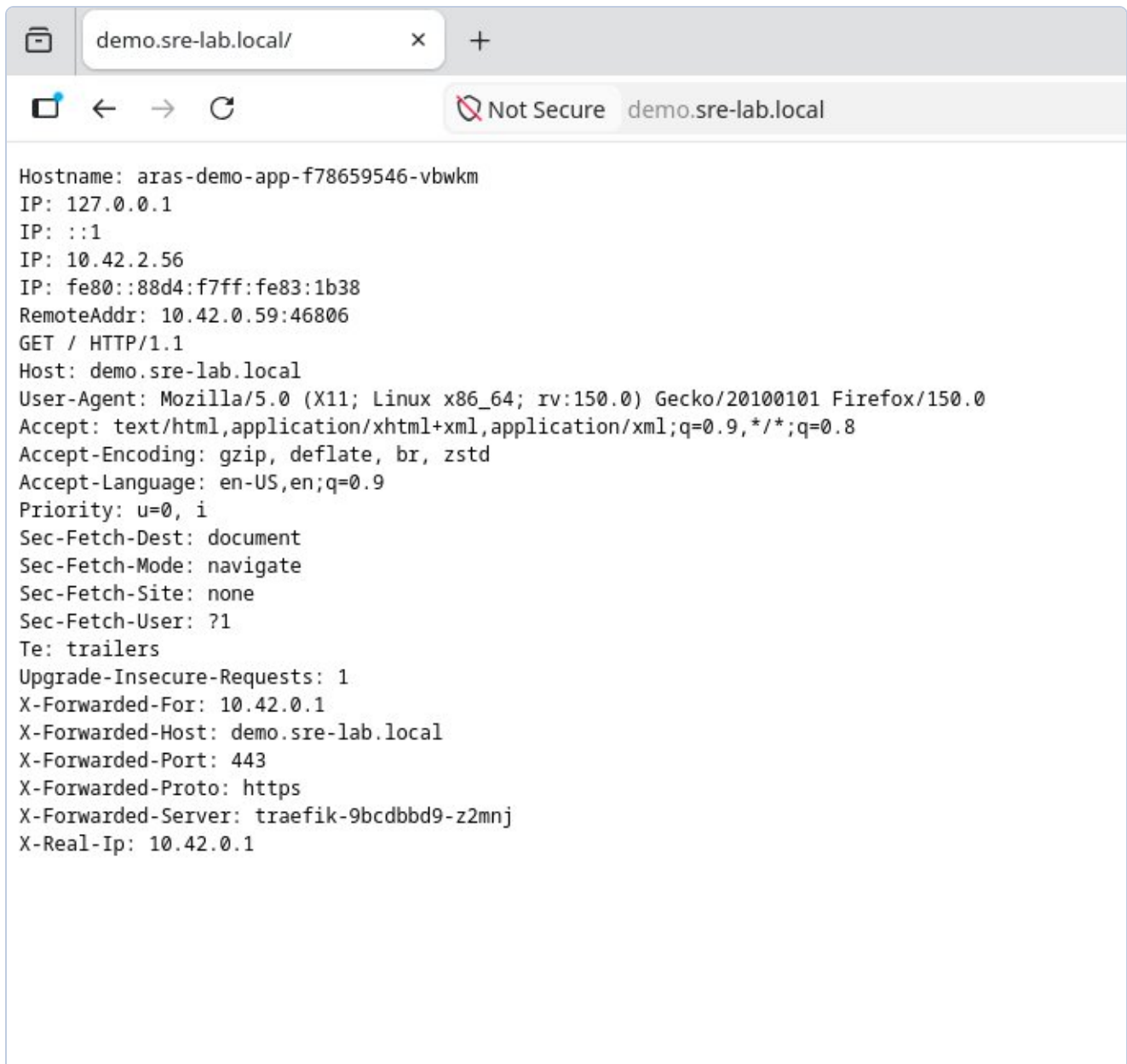
Chapter 13: Week 3 — Advanced Kubernetes Operations

WEEK 3: Advanced Kubernetes Operations

Ingress with TLS — HTTPS routing by domain

Traefik Ingress Controller (built into K3s) configured for HTTPS traffic. Generated a self-signed TLS certificate with SAN, stored it as a Kubernetes Secret, and defined routing rules: requests to demo.sre-lab.local are forwarded to the aras-demo-app service.

- Certificate: openssl x509 self-signed, SAN for demo.sre-lab.local, valid 365 days
- Secret: type kubernetes.io/tls in namespace default
- Ingress: host-based routing through Traefik on port 443
- Verification: X-Forwarded-Proto: https, X-Forwarded-Server: traefik in response



Screenshot W3-1: Firefox browser — <https://demo.sre-lab.local> working. Visible: Pod hostname, X-Forwarded-Proto: https, X-Forwarded-Server: traefik-9bcdbbd9-z2mnj, X-Forwarded-Port: 443.

HPA — CPU-based pod autoscaling

Horizontal Pod Autoscaler configured for the aras-demo-app Deployment. HPA queries metrics-server every 15 seconds for CPU usage. When average pod CPU exceeds 50% — it adds replicas. When it drops below 50% for 5 minutes — it removes pods.

- Range: minReplicas: 2, maxReplicas: 8, targetCPU: 50%
- Load test: `ab -n 50000 -c 50` — 1621 req/s over 30 seconds
- Scale-up: CPU 860% → automatic scaling to 8 replicas
- Scale-down: CPU 6% after 5 minutes → return to 2 replicas (minimum)

```
Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:32:49 2026
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS
aras-demo-hpa	Deployment/aras-demo-app	cpu: 860%/50%	2	8
REPLICAS	AGE			
4	9m55s			

aras-demo-app-5b58b48d6d-5vwl4	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-7mbln	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-859c7	1/1	Running 0	6m32s	
aras-demo-app-5b58b48d6d-96qs4	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-phz5d	1/1	Running 0	6m27s	
aras-demo-app-5b58b48d6d-sbv4q	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-xx94p	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-zrmgs	1/1	Running 0	10s	

Screenshot W3-2: HPA scale-up — cpu: 860%/50%, REPLICAS: 8 (max), 8 pods Running. Automatic scaling in action during ab load test.

```
Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:38:17 2026

NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  RE
PLICAS    AGE
aras-demo-hpa  Deployment/aras-demo-app  cpu: 6%/50%    2         8         8
15m
---
aras-demo-app-5b58b48d6d-phz5d  1/1    Running    0         11m
aras-demo-app-5b58b48d6d-zrmgs  1/1    Running    0         5m38s
```

Screenshot W3-3: HPA scale-down — cpu: 6%/50%, REPLICAS: 2 (min). After 5 minutes without traffic HPA automatically removed 6 unnecessary pods.

NetworkPolicy — intra-cluster firewall

By default all pods in Kubernetes can communicate with all pods. NetworkPolicy defines rules like a firewall — which namespaces and pods can connect to the demo application. Deployed a policy blocking all incoming traffic except from allowed sources.

- Allowed: kube-system (Traefik), monitoring (Prometheus), aras-demo-app replicas
- Blocked: everything else
- Verification: unauthorized pod — timeout/blocked. Traefik — application response

RBAC — least privilege for applications

Role-Based Access Control — every application should have its own account (ServiceAccount) with exactly the permissions it needs. Least privilege principle: if an application doesn't need to delete pods, it shouldn't be able to — even if compromised by an attacker.

- ServiceAccount: aras-demo-sa (instead of shared default)
- Role: get/list/watch on Pods and ConfigMaps — nothing more
- Verification: can-i list pods = yes, can-i delete pods = no

```
buth11@k3s-master:~$ # Test 1: nieautoryzowany Pod probuje polaczyc sie z aplikacja
# timeout 5 = czekamy max 5 sekund na odpowiedz
# Oczekiwany wynik: brak odpowiedzi / timeout - polityka blokuje
echo "=== Test: nieautoryzowany dostep (powinien byc zablokowany) ==="
sudo k3s kubectl run test-blocked \
  --image=curlimages/curl \
  --rm -it \
  --restart=Never \
  -- curl -s --max-time 5 http://aras-demo-app:8080 || echo "ZABLOKOWANY - NetworkPolicy dziala"

# Test 2: sprawdzamy ze Ingress (Traefik) nadal dziala
# Traefik jest w kube-system wiec ma dostep
echo "=== Test: dostep przez Ingress (powinien dzialac) ==="
curl -sk https://demo.sre-lab.local | head -2
=== Test: nieautoryzowany dostep (powinien byc zablokowany) ===
pod "test-blocked" deleted from default namespace
pod default/test-blocked terminated (Error)
ZABLOKOWANY - NetworkPolicy dziala
=== Test: dostep przez Ingress (powinien dzialac) ===
buth11@k3s-master:~$ # Test dostepu przez Traefik - uzywamy IP zamiast domeny
# Traefik nasluchuje na porcie 80 i 443 na wszystkich wezlach
# -H podajemy naglowek Host zeby Traefik wiedzial do ktorego serwisu kierowac
echo "=== Test dostep przez Traefik (kube-system) ==="
curl -s -H "Host: demo.sre-lab.local" http://192.168.122.10 | head -3

# Dodatkowo sprawdzamy ze bezposredni dostep do portu 8080
# przez serwis nadal dziala (LoadBalancer Service jest poza NetworkPolicy)
echo "=== Test dostep przez LoadBalancer Service ==="
curl -s http://192.168.122.10:8080 | head -3
=== Test dostep przez Traefik (kube-system) ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
=== Test dostep przez LoadBalancer Service ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
buth11@k3s-master:~$
```

Screenshot W3-4: RBAC verification — ServiceAccount aras-demo-sa: list pods=yes (allowed), delete pods=no (blocked). Least privilege principle confirmed.

Chapter 14: Postmortem POST-001

Field	Value
Severity	P2
Duration	Under 7 minutes
Root Cause	Typo: ks3-worker1 instead of k3s-worker1
Status	Resolved — regex validation in scripts

Chapter 15: Results and Conclusions

Metric	Week 1	Week 2	Week 3
Nodes	3/3 Ready	—	—
SLO	100%	—	—
Cost	\$0.00	\$0.00	\$0.00
Terraform state	Local	Remote (Blob)	—
CI/CD	—	Azure DevOps 4m24s	—
Ingress TLS	—	—	HTTPS verified
HPA	—	—	3→8→2 replicas
NetworkPolicy	—	—	Unauthorized blocked
RBAC	—	—	Least privilege
Issues	6	8	10

Chapter 16: Troubleshooting Log

ISSUE-001 KQL: CPUCapacityNanoCores column missing

project operator: Failed to resolve scalar expression named CPUCapacityNanoCores

Resolution: KubeNodeInventory | getschema — simplified query to available columns.

ISSUE-002 Azure Policy: MissingPolicyParameter cpuLimit/memoryLimit

(MissingPolicyParameter) policy assignment k8s-resource-limits is missing cpuLimit, memoryLimit

Resolution: Added --params with cpuLimit: 500m and memoryLimit: 512Mi.

ISSUE-003 Terraform: Resource Group already exists (brownfield)

Error: A resource with the ID .../SRE-Lab-RG already exists

Resolution: terraform import azurearm_resource_group.sre_lab /subscriptions/.../SRE-Lab-RG

ISSUE-004 Terraform: Saved plan is stale after import

Error: Saved plan is stale – state was changed after plan was created

Resolution: New terraform plan -out=tfplan after import.

ISSUE-005 Container Insights missing after VM restart

Resolution: az k8s-extension create --name azuremonitor-containers — agents returned in 37s.

ISSUE-006 POST-001: etcd Split-Brain (ks3 vs k3s)

FATA[0034] Node password rejected, duplicate hostname

Resolution: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent. Scripts updated with regex validation.

ISSUE-007 Azure DevOps: TerraformInstaller task missing

A task is missing: TerraformInstaller

Resolution: Install from Marketplace: ms-devlabs.custom-terraform-tasks.

ISSUE-008 Pipeline: permission for Variable Group

This pipeline needs permission to access a resource

Resolution: View → Permit — one-time authorization.

ISSUE-009 sudo does not expand ~ in paths

error: Cannot read file ~/ingress/server.crt, open ~/ingress/server.crt: no such file or directory

Root Cause: sudo runs the command as root which has a different home directory. Tilde ~ expands to /root/ instead of /home/buth11/.

Resolution: Use full path: /home/buth11/ingress/server.crt instead of ~/ingress/server.crt

ISSUE-010 HPA shows cpu: unknown — missing resource requests

```
NAME TARGETS MINPODS MAXPODS REPLICAS — cpu: <unknown>/50%
```

Root Cause: HPA calculates CPU percentage as $\text{current_usage}/\text{requests} \times 100$. Without defined requests in the Deployment, HPA has no baseline to calculate percentages from.

Resolution: Added resources.requests to Deployment: cpu: 10m, memory: 32Mi. After rolling update HPA started showing correct values: cpu: 0%/50%.